Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

Automatic Attendance System Based on Real-Time Location and Face Recognition

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CLASS: SEP490_G16

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Hanoi, June 2025

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Table 2 - List of Tables

I. Record of Changes

Date	A* M, D	In charge	Change Description
25/05/2025	A	Kiều Thị Ngọc Dung	Create document
24/07/2025	M	Kiều Thị Ngọc Dung	Update document
25/07/2025	M	Kiều Thị Ngọc Dung	Insert link Unit Test case
26/07/2025	M	Kiều Thị Ngọc Dung	Insert link Integration Test case
28/07/2025	M	Nguyễn Đình Dũng	Update Unit test, GUI test
1/08/2025	M	Nguyễn Văn Đạt	Review and update Unit test
2/08/2025	M	Kiều Thị Ngọc Dung	Create test report
03/08/2025	M	Kiều Thị Ngọc Dung	Insert link System Test case
07/08/2025	M	Kiều Thị Ngọc Dung	Update System Test case
15/08/2025	M	Kiều Thị Ngọc Dung	Update test report
16/08/2025	M	Kiều Thị Ngọc Dung	Update Acceptance test

*A - Added M - Modified D - Deleted

Table 4 - Record of changes

II. Introduction

1. System Overview

The system under development is **Real-Time Attendance System**, an advanced solution designed to automate attendance tracking using scanning data and facial recognition technology. The system is composed of three key components: a mobile application, a web-based interface, and an API backend. This solution enables organizations and events to track attendance in real time, leveraging BLE (Bluetooth Low Energy) scanning and face recognition for accurate and seamless check-ins.

Key Features:

- **Scanning Data for Attendance**: Utilizing BLE scanning and advanced algorithms to determine the precise attendance of users by detecting their proximity and verifying identity.
- **Face Recognition**: Integrating AI-based facial recognition technology to ensure that the user's identity is confirmed before marking attendance.
- **Mobile Application**: A cross-platform mobile app (primarily Android) that allows users to check in and monitor their attendance records.
- **Web Portal**: A web-based interface for administrators to manage attendance data and configure the system's settings and workflows.
- **API Integration**: Offering extensibility for future integrations with external systems such as school management or HR software.
- **Real-Time Monitoring**: Providing live updates and logs on attendance through the mobile and web platforms, ensuring accurate tracking of attendance activities.

Testing Considerations:

- Testing is essential to ensure that the system performs reliably across different environments, accurately captures scanning data, and scales effectively for larger user bases and diverse devices, especially as the system is developed iteratively.

2. Testing Strategy

2.1 Testing Types

- Testing the AAS project is an iterative process, meticulously defined in the project plan and tailored to the internal distribution of the development team. Each iteration focuses on newly developed features and customer-requested changes, ensuring that mature features are continuously and iteratively tested to prevent issues from arising. This comprehensive approach guarantees that every aspect of the project, from module code to API functions, functional levels, and the entire system, is thoroughly evaluated.

No	Type of Testing	Describe
1	**Unit Testing**	Unit testing is a type of software testing in which individual units or components of the software are tested. The purpose is to confirm that each unit of the software code works as expected.
2	**Integration Testing**	Integration testing is defined as a type of testing in which software modules are logically integrated and tested as a group. A typical software project consists of many software modules, coded by different programmers. The purpose of this level of testing is to

		reveal defects in the interaction between these software modules as they are integrated.
3	**GUI Testing**	GUI testing is a type of software testing to test the Graphical User Interface of the Software. The purpose of Graphical User Interface (GUI) Testing is to ensure that the software application functions are working according to specifications by testing the screen and controls such as menus, buttons, icons, etc. , etc.
4	**Performance Testing**	Performance testing is a software testing process used to test the speed, response time, stability, reliability, scalability, and resource usage of a software application in a particular workload. The main purpose of performance testing is to identify and remove performance bottlenecks in a software application.
5	**API Testing**	API Testing is a type of software testing that focuses on verifying that an application programming interface (API) works as expected. It checks the functionality, reliability, performance, and security of the API by sending requests and validating the responses..
6	**Regression Testing**	Regression testing is a software testing practice that ensures an application still works as expected after any code changes, updates or improvements. Regression testing is responsible for the overall stability and functionality of existing features.
7	**Data integrity testing**	Data integrity testing is a software testing process to check if the software can perform an error-free operation in a particular environment within a specified period of time. The purpose of reliability testing is to ensure that the software product is error free and reliable enough for its intended purpose.

2.2 Test Levels

Type of Test	**Test level**			
	Unit	***Integration***	***System***	***Acceptance***
Function Testing	X			
API Testing	X	X		
GUI Testing		X		X
Performance Testing	X	X		
Database Testing	X	X	X	
Regression Testing		X	X	
Data integrity testing		X		

2.3 Supporting Tools

Purpose	**Tool**	**Vendor/In-house**	**Version**
Unit Testing	Netbean, Post man	Oracle	2023
Integration Testing , System Testing	Google Excel	Google	2023
Acceptance Testing	MS Word	Microsoft	2019
Log Issue	Docs	Microsoft	2023

3. Reference Documents

No	Deliverable Item	Description
1	Unit Test	Report5.1_Unit Test
2	Integration Test	Report5.2_Integration Test
3	System Test	Report5.3_System Test
4	Acceptance Test	Acceptance Test

Table 5 - Reference Documents

III. Testing Documentation

1. Scope of Testing

Implementing quality project management according to the **Integration and Iterative model** gives professionals the assurance of faster product development and more frequent feedback. When managing projects with this model, you will enjoy the following benefits:

- **Low risk guarantee**: With iterative management and regular feedback, teams can continuously improve the product, ensuring it better matches the real needs of the market.
- **Flexibility and adaptability**: Changes are easier to accommodate during the iterative process, rather than waiting until the final stage when modifications are more costly.
- **Future-proof security**: Integrated and iterative revisions allow issues to be identified early and corrected quickly, strengthening the system's stability and security in future releases.

1.1. Testing Targets

Unit test:

- Team members will do monkey testing, check how the application handles. When performing Monkey test, members will note in the excel file the functions of the group to check if any test cases have been tested..
- Enter the number of input values to detect code weaknesses in principle:
    - If a valid input value is entered, the returned result must also be valid
    - If an invalid input value is entered, the returned result must be invalid

Integration test:

- Deploy the modules in relation to each other to see if the modules work in sync with each other or where there is a conflict.

System test:

- Function test: Make sure the product's functionality works as required, within the system's capabilities
- Database test: Checks whether the information when performing the import and export steps is the same between the database and the screen. Especially in money-related transactions, see if there are any mistakes

Acceptance test:

- Check whether the developed software product passes the acceptance standards defined on the basis of user and business requirements

System test:

● Performance test: Ensure the performance of the system under different conditions. Pagespeed to test.
● Reliability test: Ensure the system can operate for a long time without any errors.
● Usability test: Make sure the system is easy to use, easy to learn and easy to operate.
● Regression test: Ensure the stability of the system when it is integrated with different subsystems and perform different maintenance tasks.

1.2. Testing Levels

We have 4 levels of testing: Unit Test, Integration Test, System Test, Acceptance Test

2. Test Strategy

2.1 Testing Levels

No	Test Types	Purpose	Scope	Techniques
1	**Unit Testing**	To validate individual components or functions of the system, such as BLE modules, scanning algorithms, and data processing functions.	Covers small, independent modules such as:- BLE connection setup and communication.- Scanning logic for user identification.- Data format validation for both BLE and scanning systems.	- **White-box Testing**: Tests the internal logic of BLE functions, scanning algorithms, and data processing. - **Mocking**: Simulates dependencies such as the BLE device and backend API to test each module in isolation. - **Assertions**: Verifies that expected outputs from BLE and scanning modules match actual results.
2	**Integration Testing**	To ensure different system components such as BLE communication, scanning data processing, and backend API work together seamlessly.	Focuses on the interaction between:- Mobile app (UI and BLE)- BLE hardware modules and scanning components.- Backend API handling the received data.- Data sync between the app, BLE, and backend.Ensures that data flows correctly from scanning to the backend via BLE.	- **Top-Down/Bottom-Up Testing**: Tests integration layers sequentially, starting from the BLE component to the backend service.- **API Testing**: Verifies the integrity of data exchange between the mobile application and backend services using tools like Postman.- **Data Flow Testing**: Ensures that the data transmitted via BLE and scanned data is accurately processed and stored in the backend.

3	**System Testing**	To validate the system's overall functionality, ensuring end-to-end workflows such as BLE scanning and data synchronization meet the required standards.	Examines the complete workflow of the system, such as:- A user creating a scanning workflow using the mobile app. Synchronizing this workflow with the BLE module for live scanning.- Verifying data transmission from BLE to the backend for analysis.- Error logging and user feedback after scanning.	- **End-to-End Testing**: Simulates real-world usage of the system, ensuring that all components (mobile app, BLE, backend) function together as expected. **Load/Performance Testing**: Verifies the system's ability to handle large datasets, multiple BLE devices, and concurrent users.- **Stress Testing**: Simulates extreme conditions, like high data throughput, to check if the system can handle unexpected errors.
4	**Acceptance Testing**	To ensure the system meets user expectations, business requirements, and is ready for deployment. Focuses on user validation of key workflows and features such as BLE-based attendance tracking.	Focuses on validating the system's functionality from an end-user perspective, ensuring the mobile app and BLE scanning meet business goals.- Business users verify BLE scanning performance.- End-users test the accuracy and reliability of the scanning system.	- **User Acceptance Testing (UAT)**: Involves real users testing key features of the mobile app, BLE communication, and backend interaction. **Scenario Testing**: Validates user workflows such as using BLE for attendance tracking and data synchronization.- **Exploratory Testing**: Allows users to explore different scanning scenarios to ensure that unexpected user actions are handled appropriately.- **Business Process Validation**: Ensures that the BLE and scanning workflow aligns with business goals, such as ensuring accurate attendance tracking.

Table 6 - Testing Types

2.2 Test Types

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
API Testing	X	X		
Website Testing		X	X	X
Mobile Testing		X	X	X
Function Testing	X	X	X	X

Table 7 - Testing Levels

2.3 Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Testing API	Postman	Postman	Lastest
Manage test cases and data index	Google Sheet	Google	Latest
Manage bugs and the status of bugs	Microsoft Excel	Microsoft	Latest
Test Functions			Latest
Test Web Pages & Log Bugs	Google Chrome, Microsoft Edge	Google	Latest
Manage Test Case	Microsoft Excel	Microsoft	Latest
Unit Test			Latest

Table 8 - Supporting Tools

3. Test Plan

3.1 Human Resources

No	Worker/Doer	Role	Specific Responsibilities/Comments
1	Nguyễn Đình Dũng	Tester	- Create Test Plan - Review Test Case - Create Test Cases for modules - Execute test - Report test result
2	Trần Long Khánh	Developer	- Manage Test resources and assign test tasks - Execute test - Fix Bugs

3	Hà Việt Hiếu	Developer	- Create Test Cases for modules - Execute Test - Report test result - Fix Bugs
4	Nguyễn Văn Đạt	Developer	- Create Test Cases for modules - Execute Test - Report test result - Fix Bugs
5	Kiều Thị Ngọc Dung	Tester	- Create Test Plan - Review Test Case - Create Test Cases for modules - Execute test - Report test result

Table 9 - Human Resources

3.2 Test Environment

No	Purpose	Tool	Provider	Version
1	Manage test case	Excel	Microsoft	Latest
2	Write Unit Test documents	Excel	Microsoft	Latest
3	Write Test Report Documents	Excel, Word	Microsoft	Latest
4	Test API	Postman, Visual Studio	Postman Inc, Microsoft	Latest
5	Test Website	Visual Studio Code	Microsoft	Latest
6	Test Mobile	Android Studio	Jetbrains	Latest
7	Perform Unit Testing	Visual Studio, Visual Studio Code, Android Studio	Microsoft, Jetbrains	Latest
8	Perform Integration Testing	Edge, Physical Android Device	Microsoft, Google, Samsung	Latest
9	Perform System Testing	Edge, Physical Android Device	Microsoft, Google, Samsung	Latest
10	Perform Acceptance Testing	Form	Google	Latest
11	Database Integration Testing	SQL Server, Firebase	Microsoft, Firebase	Latest

Table 10 - Test Environment

3.3 Test Milestones

Milestone Task	Effort (md)	Start Date	End Date
Create all Unit Tests	10	8/7/2025	12/7/2025
Review Unit Tests	4	13/7/2025	14/7/2025
Execute all Unit Tests	4	15/7/2025	16/7/2025
Create all Integration Tests	12	17/7/2025	22/7/2025

Review Integration Tests	4	23/7/2025	24/7/2025
Execute all Integration Tests	8	24/7/2025	27/7/2025
Create all System Tests	8	27/7/2025	1/8/2025
Review System Tests	8	1/8/2025	4/7/2025
Execute all System Tests	10	5/8/2025	9/7/2025
Create Acceptance Test	2	10/8/2025	11/8/2025
Execute all Acceptance Test	8	10/8/2025	14/8/2025
Perform Acceptance Test	4	14/8/2025	15/8/2025
Total	**82**		

Table 11 - Test Milestones

4. Test Cases

In this project, we define one test case in unit testing, integration testing and system testing files must follow:

- Test case is a set of actions executed to verify a particular feature or functionality of your software application.
- The test case includes specific variables or conditions, using which a testing engineer can compare expected and actual results to determine whether a software product is functioning as per the requirements of the customer
- Unit Test Cases: UnitTest.excel
- Integration Test case: IntegrationTest.Excel
- System Test case : SystemTest.excel

5. Test Reports

5.1 Unit test

For the all result unit test : Unit Test

No	Function code	Passed	Failed	Untested	N	A	B	Total Test Cases
1	View Homepage and System Dashboard Overview(user)	6	4	0	3	4	3	10
2	Enable / Disable Notifications(user)	7	3	0	4	4	2	10
4	Authenticate via Personal Face ID(user)	10	0	0	1	6	3	10

5	Update Personal Face ID(user)	10	0	0	2	5	3	10
6	Register Personal Face ID(user)	10	0	0	2	6	2	10
7	Delete User Account (Student / Lecturer)(user)	12	0	0	3	6	3	12
8	Update User Account (Student / Lecturer)(user)	12	0	0	5	7	0	12
9	Create User Account (Student / Lecturer)(user)	10	0	0	1	8	1	10
12	log in log out(user)	10	0	0	4	6	0	10
13	View Enrollment List(schedule)	11	0	0	0	0	0	11
14	Unenroll Students from Course(schedule)	11	0	0	3	7	0	11
15	Enroll Students in Course(schedule)	10	2	0	6	6	0	12
16	Delete Schedule(shedule)	9	3	0	6	6	0	12
17	Update Schedule(scheudle)	12	0	0	3	9	0	12
18	View Schedule(schedule)	12	0	0	3	7	0	12
19	Delete Classroom(schedule)	12	0	0	4	8	0	12
20	Update Course(schedule)	14	0	0	5	9	0	14
21	View Course(schedule)	6	0	0	4	2	0	6
22	Delete Course(schedule)	14	0	0	1	13	0	14
23	Create Course(schedule)	6	0	0	2	4	0	6
25	Lecturer Synchronizes Offline Attendance Data(att)	12	0	0	6	6	0	12
26	Check Personal Attendance Record(attendance)	11	0	0	7	4	0	11
27	Student Participates in Attendance(attendance)	7	0	0	4	3	0	7
28	Student Performs Face ID Verification(attendance)	7	0	0	4	3	0	7

29	Create Face ID Verification Request(attendance)	8	0	0	2	6	0	8
30	Initialize Attendance Session(attendance)	8	0	0	1	7	0	8
32	View Registered Devices List(Device)	8	0	0	5	3	0	8
33	View Registered Devices List(Device)	8	0	0	5	3	0	8
34	Approve/Reject Device Change Request(Device)	8	0	0	3	5	0	8
35	Request Device Change(Device)	8	0	0	3	5	0	8
36	Register Device(Device)	9	0	0	4	5	0	9
37	Delete Configuration Parameter(Configuration)	7	0	0	2	5	0	7
38	Update Configuration Parameter(Configuration)	7	0	0	3	4	0	7
39	Create New Configuration Parameter(Configuration)	7	0	0	3	4	0	7
40	View Configuration Parameters(Configuration)	5	0	0	4	1	0	5
42	Student Views Personal Detailed Attendance Report(Report	6	0	0	2	4	0	6
43	Generate and View Detailed Attendance Report by Session	7	0	0	2	5	0	7
44	Generate and View Attendance Summary Report(Report)	0	0	0	0	0	0	0
48	View user notification list(Notification)	6	0	0	3	3	0	6
	Sub total	343	12	0	125	199	17	355

Table 5.1.1. Unit Test Report

	Sub total	343	12	0	125	199	17	355
	Test coverage		**100.00**	%				
	Test successful coverage		**96.62**	%				
	Normal case		**35.21**	%				
	Abnormal case		**56.06**	%				
	Boundary case		**4.79**	%				

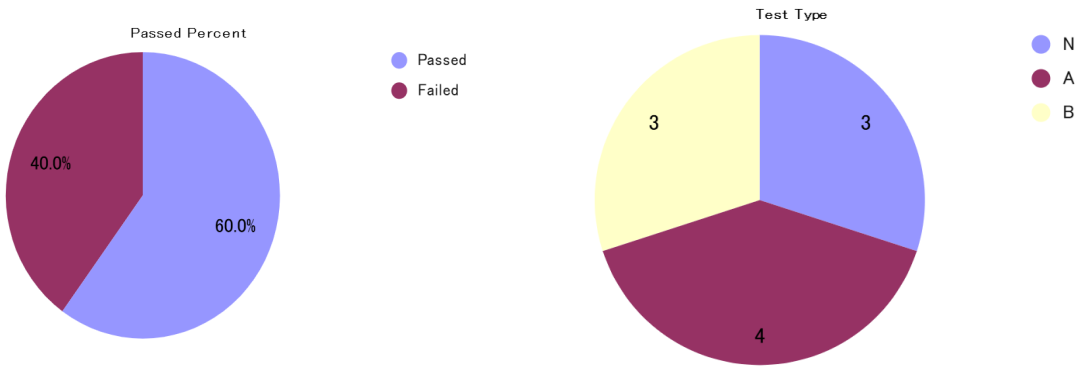


Figure 5.1.1. Unit Test Statistics

5.2 Integration Test

For the all result acceptance test : Integration Test

No	Module code	Passed	Failed	N/A	Number of test cases
1	Login/logout	9	0	0	9
2	Create User Account	10	0	0	10
3	Update User Account (Student/Lecturer)	10	0	0	10
4	Delete User Account (Student/Lecturer)	4	1	0	5
5	Register Personal Face ID	6	0	0	6
6	Update Personal Face ID	6	0	0	6
7	Authenticate using personal Face ID	7	0	0	7
8	Enable/Disable Notifications	7	0	0	7
9	View Homepage and System Dashboard	9	0	0	9
10	Create Course	9	0	0	9
11	View Course	10	0	0	10
12	Update Course	10	0	0	10

13	Delete Course	10	0	0	10
14	Create Classroom	10	0	0	10
15	View Classrooms	10	0	0	10
16	Update Classroom	10	0	0	10
17	Delete Classroom	6	0	0	6
18	Create Schedule	8	0	0	8
19	View Schedule	8	0	0	8
20	Update Schedule	6	1	0	7
21	Delete Schedule	6	2	0	8
22	Enroll Students in Course	10	0	0	10
23	Unenroll Student from Course	7	3	0	10
24	View Enrollment List	8	2	0	10
25	Initialize Attendance Session	10	0	0	10
26	Join Attendance Session	9	3	0	12
27	View Personal Attendance History	10	2	0	12
28	Student Submits Absence Report	4	2	0	6
29	Lecturer Manually Adjusts Attendance	7	0	0	7
30	Lecturer Syncs Offline Attendance Data	8	0	0	8
31	Create Face ID Verification Request	7	0	0	7
32	Device Registration	6	0	0	6
33	Device Change Request	7	0	0	7

34	Approve/Reject Device Change Request	7	0	0	7
35	View Registered Devices	8	0	0	8
36	View Configuration Parameters	6	0	0	6
37	Create New Configuration Parameter	7	0	0	7
38	Update Configuration Parameter	7	0	0	7
39	Delete Configuration Parameter	6	0	0	6
40	Generate and View Summary Attendance Report	7	0	0	7
41	Generate and View Detailed Attendance Report by Session	8	0	0	8
42	Student Views Detailed Personal Attendance Report	6	0	0	6
43	View Configuration Parameters	8	0	0	8
	Sub total	334	16	0	350
	Test coverage		**100.00**	%	
	Test successful coverage		**95.43**	%	

Table 5.2.1. Integration Test Report

5.3 System Test

For the full test case list refer system_Test

No	Module code	Passed	Failed	N/A	Number of test cases
1	View Homepage and System Dashboard(USER)	10	0	0	10

2	Enable/Disable Notifications(USER)	9	1	0	10
3	Authenticate using Face ID(USER)	9	1	0	10
4	Update Personal Face ID(USER)	9	0	0	9
5	Register Personal Face ID(USER)	8	0	0	8
6	Delete User Account (Student/Lecturer)(USER)	9	1	0	10
7	Update User Account (Student/Lecturer)(user)	16	0	0	16
8	User Account Creation(user)	10	1	0	11
9	Login/Logout(user)	14	0	0	14
10	View Enrollment List(Schedule)	15	0	0	15
11	Unenroll Students from Course(Schedule)	13	2	0	15
12	Delete Schedule(Schedule)	15	0	0	15
13	Update Schedule(Schedule)	18	0	0	18
14	View Schedule(Schedule)	18	0	0	18
15	Create Schedule (Manual & CSV/Excel Import)(Schedule)	18	0	0	18
16	Delete Classroom(Schedule)	10	0	0	10
17	Update Classroom(Schedule)	12	0	0	12
18	Create Classroom(Schedule)	12	0	0	12
19	Delete Course(Schedule)	10	0	0	10
20	Update Course(Schedule)	10	0	0	10
21	View Courses(Schedule)	10	0	0	10
22	Create Course(Schedule)	12	0	0	12
23	Create Face ID Verification Request(Attendance)	10	0	0	10
24	Lecturer Syncs Offline Attendance Data(Attendance)	15	0	0	15
25	Lecturer Manually Adjusts Attendance(Attendance)	15	0	0	15

26	Student Submits Absence Report(Attendance)	13	2	0	15
27	View Personal Attendance History(Attendance)	15	0	0	15
28	Join Attendance Session(Attendance)	15	0	0	15
29	Initiate Attendance Session(Attendance)	11	0	0	11
30	View Registered Device List(device)	12	0	0	12
31	Request Device Change(device)	11	0	0	11
32	Register Device(device)	10	0	0	10
33	View User Notification List(Notification)	13	0	0	13
34	UC: Student Views Detailed Personal Attendance Report	11	0	0	11
35	UC: Generate and View Detailed Attendance Report by Session	8	0	0	8
36	UC: Generate and View Summary Attendance Report(Reporting)	12	0	0	12
37	Delete Configuration Parameter(Configuration)	9	1	0	10
38	Update Configuration Parameter(Configuration)	11	2	0	13
39	View Configuration Parameters(Configuration)	11	1	0	12
40	Create New Configuration Parameter(Configuration)	12	1	0	13
	Sub total	481	13	0	494
	Test coverage		**100.00**	%	
	Test successful coverage		**97.37**	%	

Table 5.3.1. System Test Report

5.4 Acceptance Test

For the all result acceptance test : Acceptance Testing (Responses)

No	Question	Yes	No
1	In the dashboard, is there enough necessary information?	V	
2	Do the functions work correctly?	V	
3	Is the app responsive?	V	

4	Report is easy to use?	V	
5	Are the messages clear and easy to use?	V	
6	Are all the dates formatted?	V	
7	Are there enough reports for work?		V
8	Do you have any difficulty in using this system?		V